Akoizo Original Practice - Form 1

Division B | Rocks And Minerals | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Mohs hardness is…

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

A. An ordinal scratch scale

B. A linear mass scale

C. A color scale

D. A toughness scale

Answer: _____

2. (2 points) Pyrite belongs to…

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

A. Sulfides

B. Halides

C. Carbonates

D. Native elements

Answer: _____

3. (2 points) Metamorphism usually involves…

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

A. Solid-state change

B. Complete melting by definition

C. Only loose sediment

D. No chemical reactions

Answer: _____

4. (2 points) Same composition, different structure defines…

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

A. Sorting

B. Streak

C. Cementation

D. Polymorphism

Answer: _____

5. (2 points) Fractional crystallization requires…

Unit: Igneous processes, textures, and Bowen's series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

A. Only a color change

B. No crystals

C. Sediment compaction

D. Separation of crystals from melt

Answer: _____

6. (2 points) An error log should record…

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

A. Only total score

B. Only elapsed time

C. Only the answer letter

D. The failed reasoning and correction

Answer: _____

7. (2 points) Quartz is a…

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

A. Sulfide

B. Halide

C. Framework silicate

D. Double-chain silicate

Answer: _____

8. (2 points) Sorting describes…

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

A. Mineral hardness

B. Fossil age

C. Spread of grain sizes

D. Edge smoothness

Answer: _____

9. (4 points) Why can streak fail on quartz?

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

Answer: _____
_____
_____

10. (4 points) Why is luster insufficient for gold identification?

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

Answer: _____
_____
_____

11. (4 points) Compare contact and regional metamorphism.

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

Answer: _____
_____
_____

12. (4 points) What is a pseudomorph?

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

Answer: _____
_____
_____

13. (4 points) What do vesicles record?

Unit: Igneous processes, textures, and Bowen's series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

Answer: _____
_____
_____

14. (4 points) Why use multiple images?

Unit: Identification strategy, references, and timed practice.
Worked-case data: You have 60 seconds and two candidate minerals. Plan an evidence-first response.

Answer: _____
_____
_____

15. (4 points) Explain oxygen sharing.

Unit: Silicate structures and mineral groups.
Worked-case data: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

Answer: _____
_____
_____

16. (4 points) Give two preservation factors.

Unit: Sedimentary processes, grains, and fossils.
Worked-case data: A rock contains rounded gravel clasts in cement. Classify and constrain the claim.

Answer: _____
_____
_____

17. (4 points) Name one alternative-evidence check.

Unit: Minerals, rocks, and diagnostic evidence.
Worked-case data: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

Answer: _____
_____
_____

18. (4 points) Explain a mineral use through a property.

Unit: Non-silicates and economic geology.
Worked-case data: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

Answer: _____
_____
_____